Advanced OPERATIVE DENTISTRY

Commissioning Editor: Alison Taylor
Development Editor: Carole McMurray
Project Manager: Kiruthiga Kasthuriswamy
Designer: Kirsteen Wright
Illustration Manager: Bruce Hogarth

Advanced OPERATIVE DENTISTRY
A PRACTICAL APPROACH

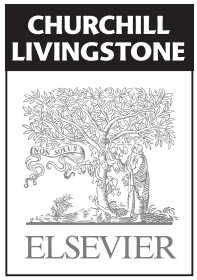
Edited by

Professor David Ricketts

Professor of Cariology and Conservative Dentistry/Honorary Consultant in Restorative Dentistry
Leader of the Section of Operative Dentistry, Fixed Prosthodontics and Endodontology,
Dundee Dental School, University of Dundee, UK

Professor David Bartlett

Professor of Prosthodontics/Honorary Consultant in Restorative Dentistry
Head of Prosthodontics, Kings College London Dental Institute, Guy's Hospital, London, UK



Edinburgh London New York Oxford Philadelphia St Louis Sydney Toronto 2011

ELSEVIER
CHURCHILL
LIVINGSTONE

© 2011 Elsevier Ltd. All rights reserved.

No part of this publication may be reproduced or transmitted in any form or by any means, electronic or mechanical, including photocopying, recording, or any information storage and retrieval system, without permission in writing from the publisher. Details on how to seek permission, further information about the Publisher's permissions policies and our arrangements with organizations such as the Copyright Clearance Center and the Copyright Licensing Agency, can be found at our website: http://www.elsevier.com/permissions.

This book and the individual contributions contained in it are protected under copyright by the Publisher (other than as may be noted herein).

ISBN 978 0 7020 3126 7

British Library Cataloguing in Publication Data
A catalogue record for this book is available from the British Library

Library of Congress Cataloging in Publication Data
A catalog record for this book is available from the Library of Congress

Notices
Knowledge and best practice in this field are constantly changing. As new research and experience broaden our understanding, changes in research methods, professional practices, or medical treatment may become necessary.

Practitioners and researchers must always rely on their own experience and knowledge in evaluating and using any information, methods, compounds, or experiments described herein. In using such information or methods they should be mindful of their own safety and the safety of others, including parties for whom they have a professional responsibility.

With respect to any drug or pharmaceutical products identified, readers are advised to check the most current information provided (i) on procedures featured or (ii) by the manufacturer of each product to be administered, to verify the recommended dose or formula, the method and duration of administration, and contraindications. It is the responsibility of practitioners, relying on their own experience and knowledge of their patients, to make diagnoses, to determine dosages and the best treatment for each individual patient, and to take all appropriate safety precautions.

To the fullest extent of the law, neither the Publisher nor the authors, contributors, or editors, assume any liability for any injury and/or damage to persons or property as a matter of products liability, negligence or otherwise, or from any use or operation of any methods, products, instructions, or ideas contained in the material herein.

ELSEVIER your source for books, journals and multimedia in the health sciences
www.elsevierhealth.com

Working together to grow libraries in developing countries
www.elsevier.com | www.bookaid.org | www.sabre.org
ELSEVIER BOOK AID International Sabre Foundation

The Publisher's policy is to use **paper manufactured from sustainable forests**

Printed in China

Contributors

Professor David Bartlett
Professor of Prosthodontics/Honorary Consultant in Restorative Dentistry,
Head of Prosthodontics, Kings College London Dental Institute, Guy's Hospital, London, UK

Professor David Ricketts
Professor of Cariology and Conservative Dentistry/Honorary Consultant in Restorative Dentistry,
Section Leader,
Section of Operative Dentistry, Fixed Prosthodontics and Endodontology,
Dundee Dental School,
Dundee, UK

Dr Graham Chadwick
Senior Lecturer/Honorary Consultant in Restorative Dentistry,
Dundee Dental School,
Dundee, UK

Dr Angela Gilbert
Senior Lecturer in Restorative Dentistry,
Dundee Dental School,
Dundee, UK

Dr Andrew Hall
Senior Lecturer/Honorary Consultant in Restorative Dentistry,
Dundee Dental School,
Dundee, UK

Dr John Radford
Senior Lecturer/Honorary Consultant in Restorative Dentistry,
Dundee Dental School,
Dundee, UK

Professor William Saunders
Professor of Endodontology/Honorary Consultant in Restorative Dentistry,
Dundee Dental School,
Dundee, UK

Dr Brian Stevenson
Lecturer/Honorary Specialist Registrar in Restorative Dentistry,
Dundee Dental School,
Dundee, UK

Dr Carol Tait
Senior Clinical Teaching Fellow,
Dundee Dental School,
Dundee, UK

Preface

Advanced operative procedures in dentistry encompass all dental subjects and disciplines such as oral surgery, the placement and restoration of dental implants and endodontics to name a few; however, these subjects are comprehensively addressed in other texts. This book therefore concentrates on fixed prosthodontics which involves the preparation of teeth for laboratory-made indirect restorations such as crowns, bridges, veneers, inlays and onlays. Where alternative, more conservative treatments are possible – for example the placement of direct composite to alter the shape of teeth – these are also described.

The necessity for advanced indirect restorations is the result of disease or conditions which have compromised the dentition. It is important to appreciate that using restorations does not treat the diseases or conditions, and their identification and prevention underpin successful treatment. The first five chapters of this book therefore focus on the main diseases and conditions (dental caries, periodontal disease, endodontic problems, tooth wear and aesthetic problems) which can lead to the need for advanced operative techniques and addresses how these techniques can impact on the remaining dentition and oral health. In the ensuing chapters details of materials, techniques and tooth preparation are described which aim to empower the reader to achieve a high standard of care for their patients.

Indirect restorations are made in a dental laboratory and clear communication between dentist and laboratory technician is essential. This takes the form of accurate impressions, occlusal registration and aspects of appearance such as the shade and contour required, all of which are devoted an individual chapter. Clear prescription of restoration design and material choice is also required and these are discussed in relation to the individual restorations. In many dental schools little or no laboratory work is carried out by undergraduate dental students; however, an understanding of how indirect restorations are made is important as certain aspects of tooth preparation have to be followed to facilitate laboratory construction. Laboratory procedures can also impact upon how materials and restorations are handled at the chairside. As such, these aspects of laboratory work are described throughout the book.

The provision of advanced indirect restorations is costly and can often have an impact on the remaining dentition and dental health. The advantages and disadvantages associated with their provision have to be balanced for each individual patient for them to be able to give informed consent. These principles underlie successful treatment planning, execution and hence patient care, and are the principles behind this text.

Acknowledgements

This textbook would not have been possible if it were not for the patients who allowed their clinical photographs to be used for medical education; the authors, undergraduate students, postgraduate students and clinical trainees who allowed their clinical work to be included; the dental technicians who made the indirect restorations used to illustrate this book, in particular John McLeish, Kevin Linklater, Donald Aitkenhead, Stewart Fairlie and Dave McMahon; and clinical photographer Simon Scott whose expertise in clinical and laboratory imaging has been invaluable. The authors would like to sincerely thank all those who have been involved and given their time generously and unreservedly.